

Lab 4

Pipelining and Datapath



Task	Objective Description
Task 1: Behavioural Simulation	To examine the uP16 pipelined processor structure, modify instruction memory contents, synthesize the design, and verify correct operation using behavioural simulation. (<u>I have only explained this in video</u>)
Task 2: Post-Synthesis Timing Simulation	To perform timing simulation and compare it with behavioural simulation to observe propagation delay effects.
Task 3: FPGA Implementation	To implement the uP16 processor on the Basys 3 board and verify correct program execution via 7-segment display and LEDs.



16 bit microprocessor- no data forwarding

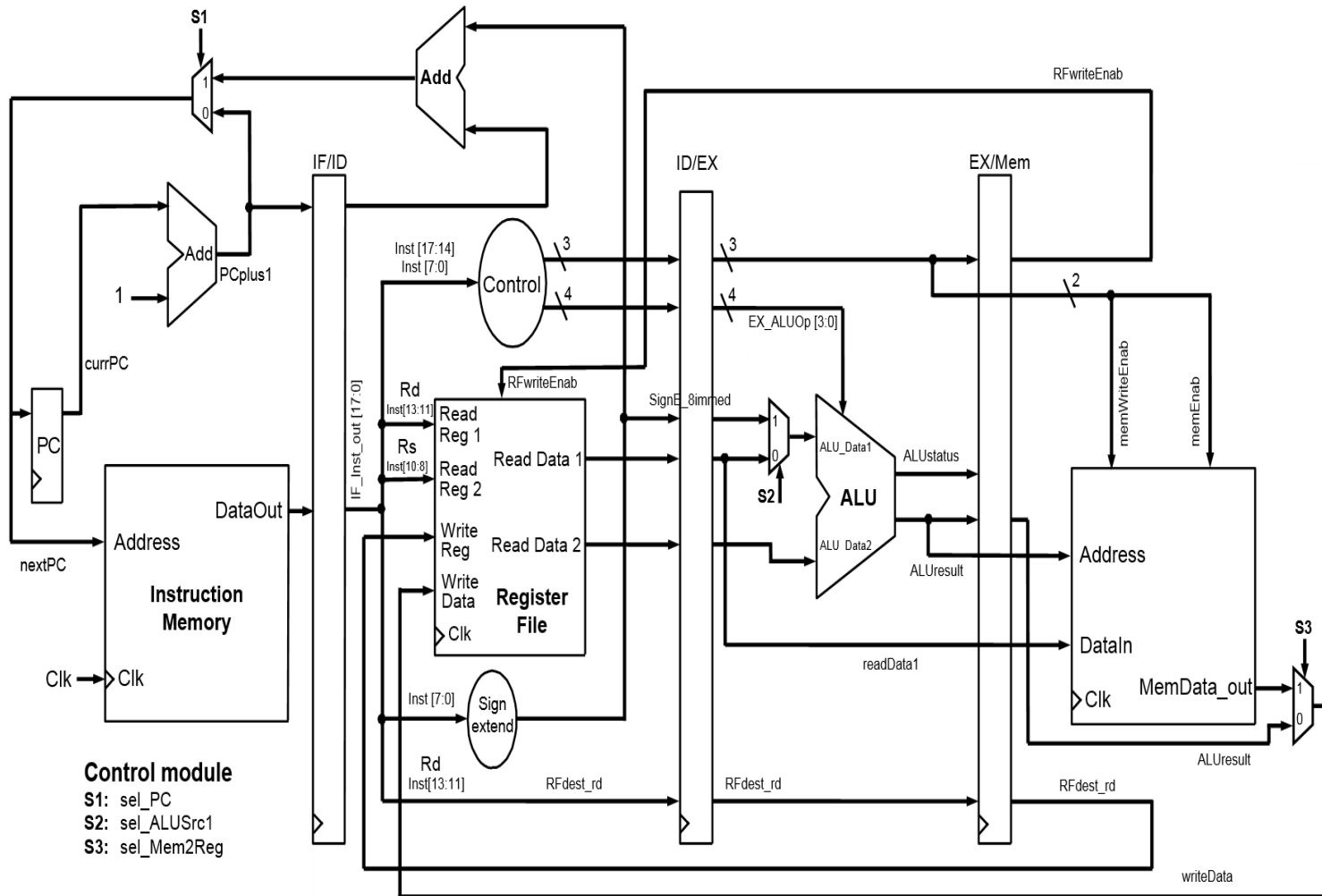


Table 1: The 18 bit instruction formats for the uP16 processor

Field	4 bits	3 bits	3 bits	8 bits
R-format	OpCode	Rd	Rs	Function/Immed

Table 2: The instruction set for the uP16 processor

Name	op	Funct/Immed	Instruction	Effect(s)
nop	0x00	0x00	nop	Do nothing (instruction word = 0)
add	0x00	0x01	add rd, rs	$RF[rd] = RF[rd] + RF[rs];$
sub	0x00	0x02	sub rd, rs	$RF[rd] = RF[rd] - RF[rs];$
and	0x00	0x03	and rd, rs	$RF[rd] = RF[rd] \text{ and } RF[rs];$
or	0x00	0x04	or rd, rs	$RF[rd] = RF[rd] \text{ or } RF[rs];$
nand	0x00	0x05	nand rd, rs	$RF[rd] = RF[rd] \text{ nand } RF[rs];$
nor	0x00	0x06	nor rd, rs	$RF[rd] = RF[rd] \text{ nor } RF[rs];$
xor	0x00	0x07	xor rd, rs	$RF[rd] = RF[rd] \text{ xor } RF[rs];$
not	0x00	0x08	not rd, rs	$RF[rd] = \text{not } RF[rs];$
mov	0x00	0x0A	mov rd, rs	$RF[rd] = RF[rs];$
lw	0x01	--	lw rd, rs, immed	$RF[rd] \leftarrow \text{memory}(RF[rs] + \text{immed});$
sw	0x02	--	sw rd, rs, immed	$RF[rd] \rightarrow \text{memory}(RF[rs] + \text{immed});$
lli	0x03	immed	lli rd, rs, immed	$RF[rd] = \text{sign_ext}(\text{immed});$ (rs is unused)
lui	0x04	immed	lui rd, rs, immed	$RF[rd] = \{\text{immed}, RF[rs][7:0]\}$
addi	0x05	immed	addi rd, rs, immed	$RF[rd] = RF[rs] + \text{sign_ext}(\text{immed});$
	0x6-8			Reserved
beq	0x08	immed	beq rd, rs, immed	if $(RF[rd] == RF[rs])$ pc += sign_ext(immed) + 1;
bne	0x09	immed	bne rd, rs, immed	if $(RF[rd] != RF[rs])$ pc += sign_ext(immed) + 1;
blt	0x0A	immed	bltz rd, rs, immed	if $(RF[rd] < RF[rs])$ pc += sign_ext(immed) + 1;
bgt	0x0B	immed	bgtz rd, rs, immed	if $(RF[rd] > RF[rs])$ pc += sign_ext(immed) + 1;
	0xC-F			Reserved

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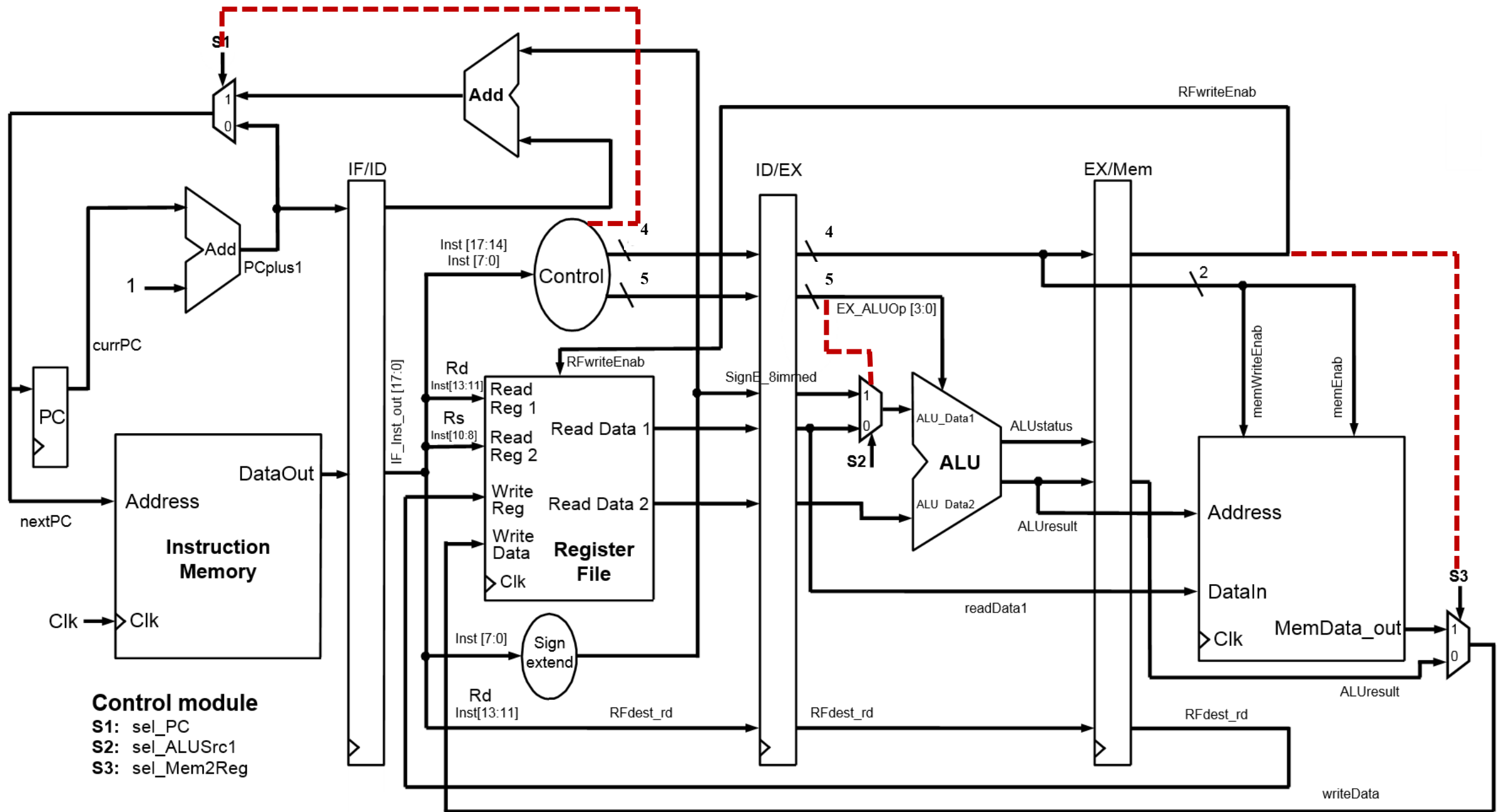
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	0xC-F			Reserved

For add/sub/and/or/nand/nor/xor/not and mov- opcode is 0x00 (shown in red) , hence ALUop from control logic uses the function field (in orange)

Inst	ALUop	sel_ALU src1	mem2 Reg	mem Enab	memWrite Enab	RFwrite Enab	PC_select
LW	0001	1	1	1	0	1	0
SW	0001	1	0	1	1	0	0
LLI	1011	1	0	0	0	1	0
LUI	1100	1	0	0	0	1	0
ADDI	0001	1	0	0	0	1	0
BEQ	X	X	0	0	0	0	1(readData1 == readData2)
BNE	X	X	0	0	0	0	1(readData1 != readData2)
BLT	X	X	0	0	0	0	1(readData1 < readData2)
BGT	X	X	0	0	0	0	1(readData1 > readData2)
default	0000	X	0	0	0	0	0



Hierarchy after including all files for 16 bit processor

- ▼ ● **uP16_top** (uP16_top.v) (7)
 - > ● T1 : IF_stage (IF_stage.v) (1)
 - T2 : IF_ID_PR (IF_ID_PR.v)
 - > ● T3 : ID_stage (ID_stage.v) (2)
 - T4 : ID_EX_PR (ID_EX_PR.v)
 - > ● T5 : EX_stage (EX_stage.v) (1)
 - T6 : EX_Mem_PR (EX_Mem_PR.v)
 - > ● T7 : Mem_stage (Mem_stage.v) (1)

The screenshot shows the 'Sources' window of a design tool. The title bar is 'Sources'. Below the title bar is a toolbar with icons for search, zoom in, zoom out, add, help, and a counter showing '0'. The main area displays a tree view of the design sources. The root is 'Design Sources (1)'. Under it is 'uP16_top (uP16_top.v) (7)'. This module contains several sub-modules: 'T1 : IF_stage (IF_stage.v) (1)' which contains 'IF_1 : I_mem (Instruct_mem.v)'; 'T2 : IF_ID_PR (IF_ID_PR.v)'; 'T3 : ID_stage (ID_stage.v) (2)' which contains 'EX_1 : Reg_File (ID_stage.v)' and 'EX_3 : Control (ID_stage.v)'; 'T4 : ID_EX_PR (ID_EX_PR.v)'; 'T5 : EX_stage (EX_stage.v) (1)' which contains 'EX_1 : ALU (EX_stage.v)'; 'T6 : EX_Mem_PR (EX_Mem_PR.v)'; and 'T7 : Mem_stage (Mem_stage.v) (1)' which contains 'DM_1 : D_mem (Data_mem.v)'.

Sources

▼ Design Sources (1)

- ▼ ● **uP16_top** (uP16_top.v) (7)
 - ▼ ● T1 : IF_stage (IF_stage.v) (1)
 - IF_1 : I_mem (Instruct_mem.v)
 - T2 : IF_ID_PR (IF_ID_PR.v)
 - ▼ ● T3 : ID_stage (ID_stage.v) (2)
 - EX_1 : Reg_File (ID_stage.v)
 - EX_3 : Control (ID_stage.v)
 - T4 : ID_EX_PR (ID_EX_PR.v)
 - ▼ ● T5 : EX_stage (EX_stage.v) (1)
 - EX_1 : ALU (EX_stage.v)
 - T6 : EX_Mem_PR (EX_Mem_PR.v)
 - ▼ ● T7 : Mem_stage (Mem_stage.v) (1)
 - DM_1 : D_mem (Data_mem.v)

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1	2				<u>OP CODE</u>	<u>R[rd]</u>	<u>R[rs]</u>	<u>D/ITRG</u>		<u>Assembly Code</u>		<u>Code</u>	<u>Parity</u>	<u>Data</u>		<u>RAM Data Init</u>
2	00	>>			nop					nop		00000		0000		
3	01	>>			lli	r4		71		lli r4, #71		0E071	0	E071		
4	02	>>			nop					nop		00000		0000		
5	03	>>			lli	r1		1		lli r1, #1		0C801	0	C801		
6	04	>>			lli	r2		2		lli r2, #2		0D002		D002		
7	05	>>			lui	r4	r4	f9		lui r4, r4, #f9		124F9	4	24F9		
8	06	>>			add	r2	r1			add r2, r1		01101		1101		
9	07	>>			mov	r5	r3			mov r5, r3		0280A	0	280A		
10	08	>>			addi	r3	r1	1		addi r3, r1, #1		15901		5901		
11	09	>>			addi	r6	r4	5		addi r6, r4, #5		17405	5	7405		
12	0A	>>			beq	r0	r0	4		beq r0, r0, #4		20004		0004		
13	0B	>>			nop					nop		00000	2	0000		
14	0C	>>			add	r4	r1			add r4, r1		02101		2101		
15	0D	>>			or	r2	r4			or r2, r4		01404	0	1404		
16	0E	>>			nop					nop		00000		0000		
17	0F	>>			sw	r4	r1	0		sw r4, r1, #0		0A100	0	A100	00	.INIT_00(256'hA100_0000_1404_2101_0000_0004_7405_5901_2B0A_1101_24F9_D002_C801_0000_0000_0000_0000)
18	10	>>			sw	r5	r1	1		sw r5, r1, #1		0A901		A901		
19	11	>>			nop					nop		00000	0	0000		
20	12	>>			lw	r6	r1	0		lw r6, r1, #0		07100		7100		


```

.DO_REG(0), // Optional output register (0 or 1)
//.INIT(36'h000000000), // Initial values on output port
.INIT_FILE ("NONE"),
.WRITE_WIDTH(18), // Valid values are 1-72 (37-72 only valid when BRAM_SIZE="36Kb")
.READ_WIDTH(18), // Valid values are 1-72 (37-72 only valid when BRAM_SIZE="36Kb")
//.SRVAL(36'h000000000), // Set/Reset value for port output
.WRITE_MODE("WRITE_FIRST"), // "WRITE_FIRST", "READ_FIRST", or "NO_CHANGE"
.INIT_00(256'hA100_0000_1404_2101_0000_0004_7405_5901_2B0A_1101_24F9_D002_C801_0000_E071_0000),
.INIT_01(256'h0000_0000_1105_0A01_0000_5806_3202_D800_0000_2B01_1901_7901_0E01_7100_0000_A901),
.INIT_02(256'h0000_0000_0000_0000_0000_0000_0000_0000_0000_0000_0000_0000_0000_0000_00DE_0C08_0000),
.INIT_03(256'h0000000000000000000000000000000000000000000000000000000000000000),
.INIT_04(256'h0000000000000000000000000000000000000000000000000000000000000000),
.INIT_05(256'h0000000000000000000000000000000000000000000000000000000000000000),
.INIT_06(256'h0000000000000000000000000000000000000000000000000000000000000000).

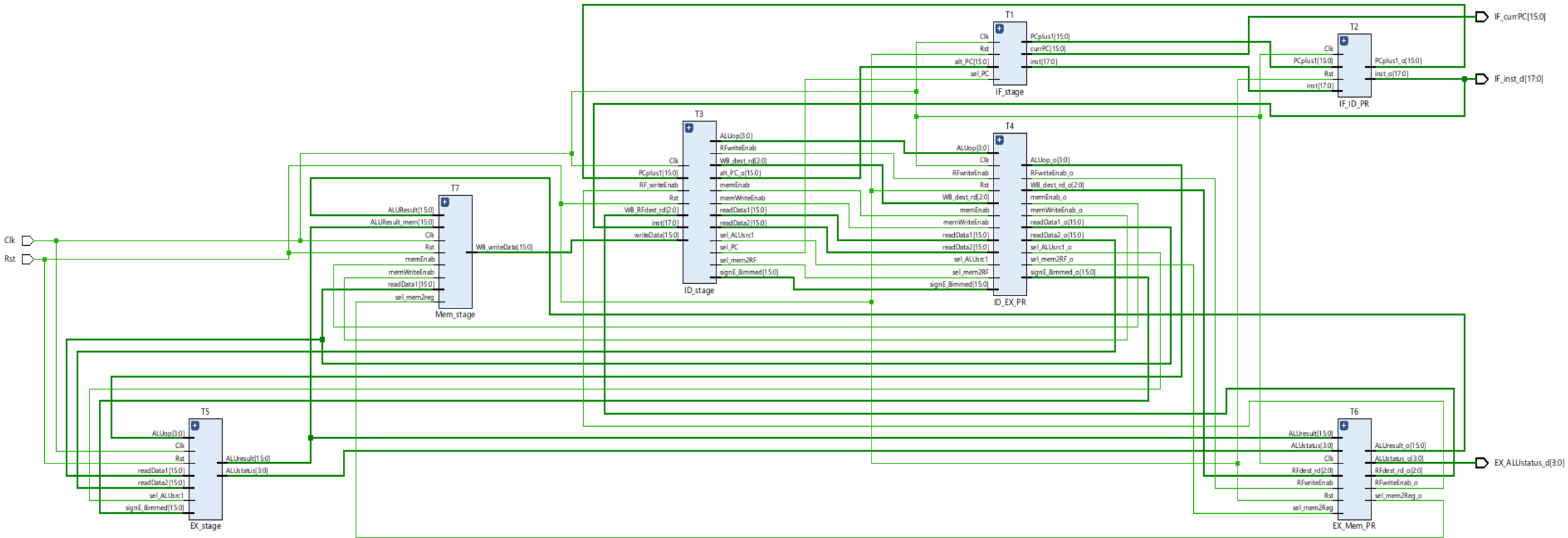
// The next set of INITP_xx are for the parity bits
.INITP_00(256'h00000000000000000000000000000000000000000000000000000000200020000000250400),
.INITP_01(256'h0000000000000000000000000000000000000000000000000000000000000000),
.INITP_02(256'h0000000000000000000000000000000000000000000000000000000000000000),
.INITP_03(256'h0000000000000000000000000000000000000000000000000000000000000000),
.....

```

From uP16_Assembler.xlsx,
take the codes for instruction
and parity and paste inside the
.INIT and .INITP

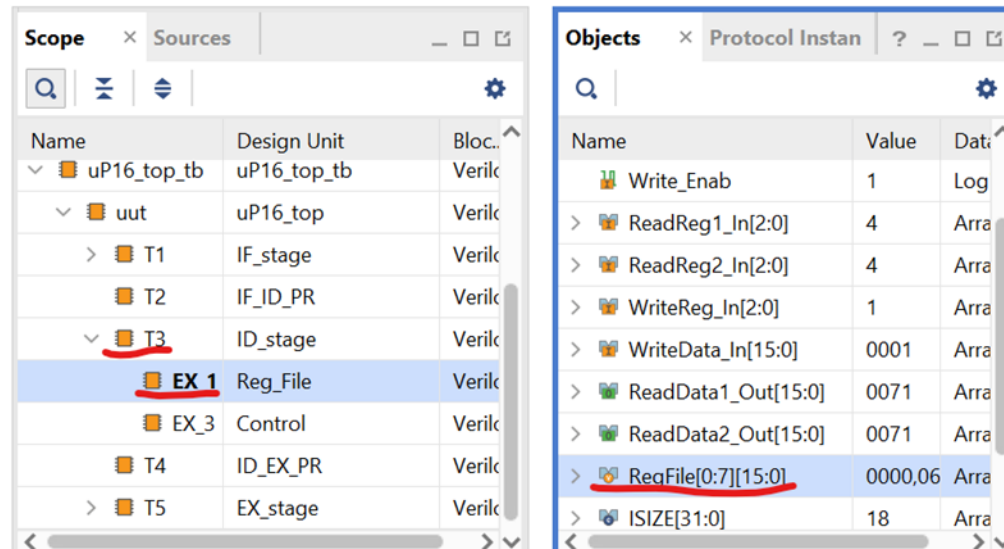
Note that 2nd instruction is still
NOP and not changed to lli r3,ff

16 bit microprocessor- no data forwarding- RTL schematic



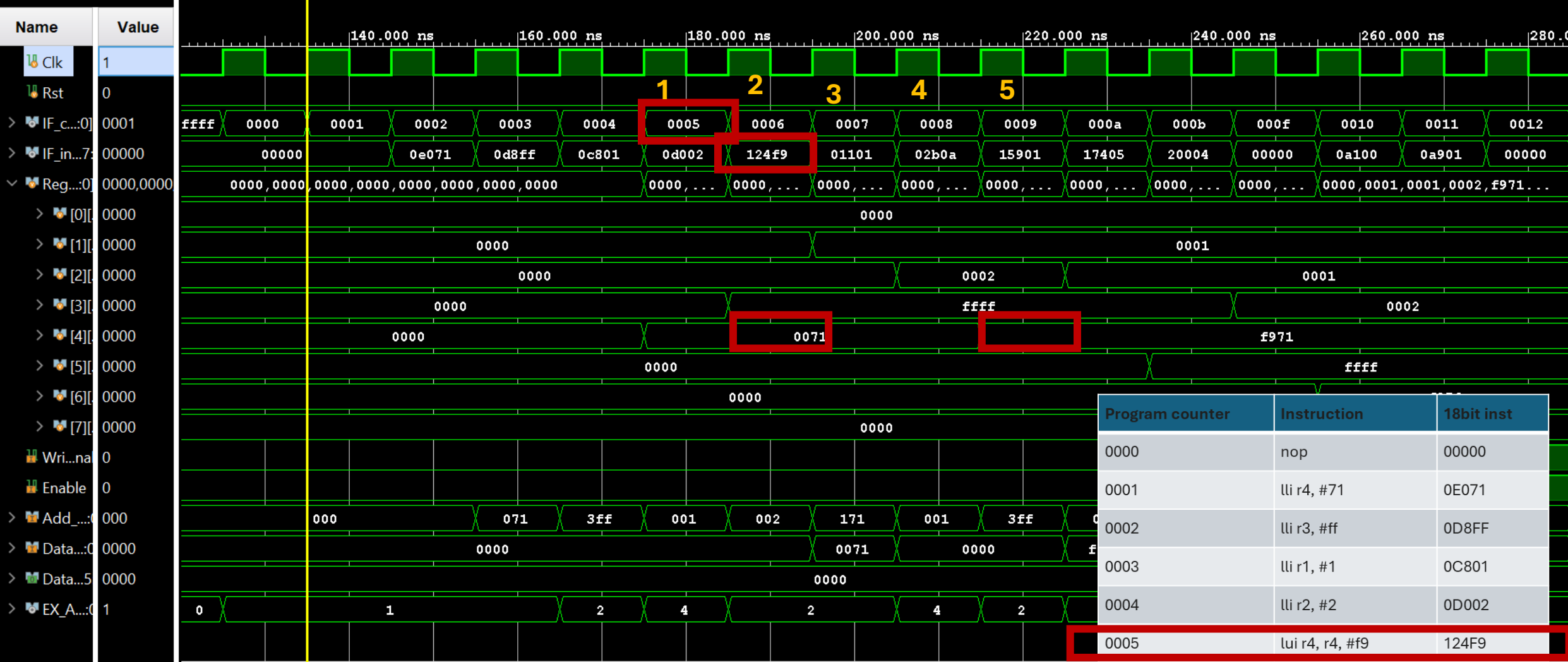
Task 1- Simulation using test bench (no propagation delay)

- After Synthesis, we need to do add the test bench as a simulation source and run the simulation
- Add the register file (RF), *RegFile* [0:7,15:0], and Data memory, Write_Enab, Enable, Add_In, Data_in, Data_out simulation objects to the simulation.
- To do this open the *uP16_top_tb* instance in the “Scope” pane. Open *uut*, and select the sub-instance (the various instances match the module instantiation instances in *uP_top.v*) until you have the *RegFile* element. Then drag the simulation object (from the ‘Objects” pane) to the simulation “Name” pane. Repeat for rest.

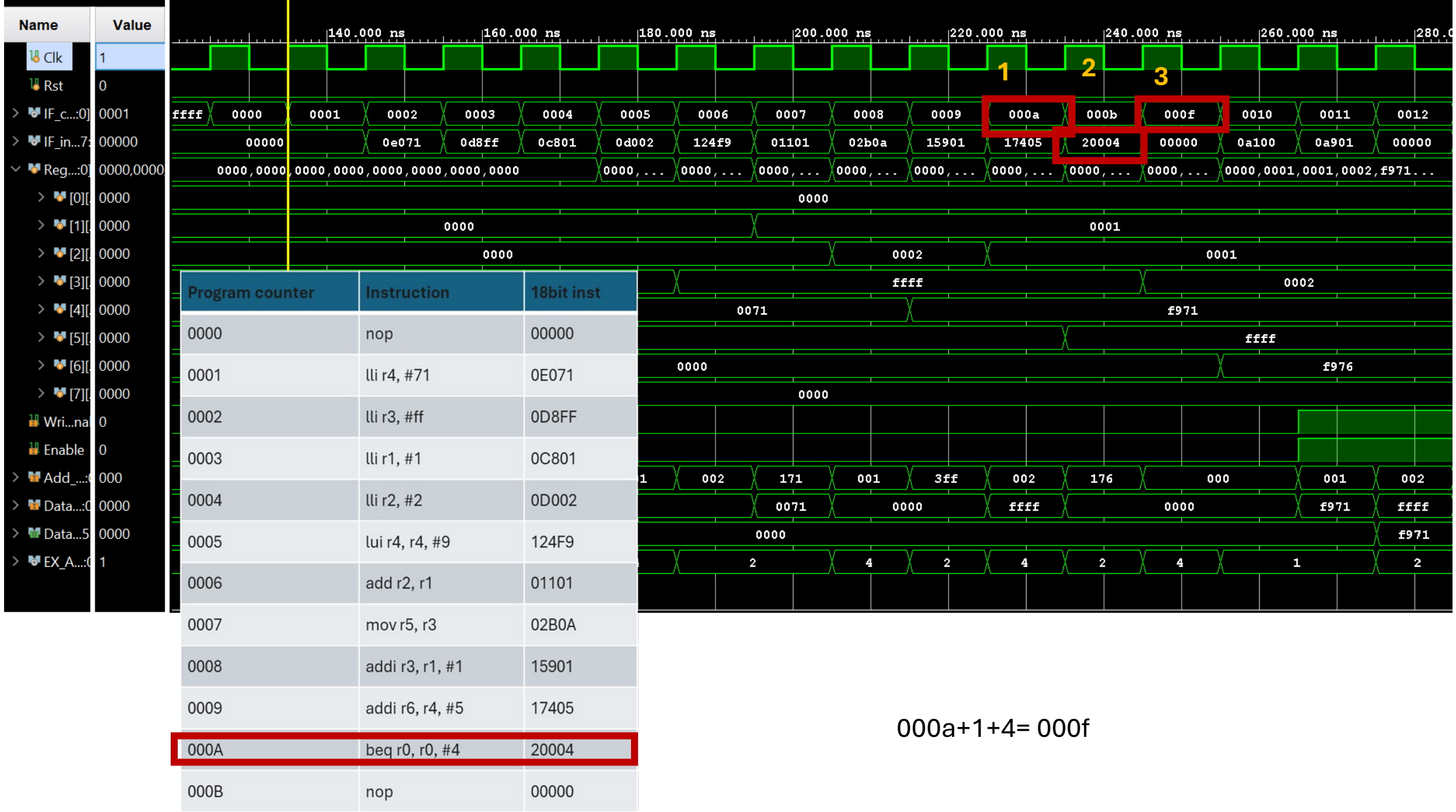








Program counter	Instruction	18bit inst
0000	nop	00000
0001	lli r4, #71	0E071
0002	lli r3, #ff	0D8FF
0003	lli r1, #1	0C801
0004	lli r2, #2	0D002
0005	lui r4, #f9	124F9
0006	add r2, r1	01101
0007	mov r5, r3	02B0A
0008	addi r3, r1, #1	15901
0009	addi r6, r4, #5	17405
000A	beq r0, r0, #4	20004
000B	nop	00000





(Task 1) Simulation window with upation
of register file as explained earlier



(Task 2) Same simulation window with
propagation delay shown



(Task3) Implementation uP16 design
on the Basys3 board